

Recovering latent hierarchy without gradient descent: a negative result for transformers, a verified estimator, and calibrated instruments for detection

Abstract

We study recovery of the latent symbols of a depth-4 probabilistic context-free grammar from its leaf strings. Three results. First, across twelve transformer configurations—varying depth, width, input representation, output factorisation, architecture family, auxiliary supervision and curriculum—a linear probe for the level-3 latent symbol never leaves its shuffled-label floor, although an exact computation shows 1.87 nats of level-3 information are present in the observed prefix. Second, a gradient-free estimator recovers the grammar to 0.001 nats per sequence of the true model in 131 seconds, once two things are fixed: the regime must be identifiable (we give the observations-per-parameter arithmetic separating 0.15 from 781), and the clustering must happen *inside* EM rather than being imposed beforehand. A hard partition of observed child pairs cannot represent a grammar in which pairs are shared across parents, which 4 of 16 level-1 pairs are here; split-merge moves scored by Bregman information close the remaining gap. Third, we build and validate instruments for detecting hierarchy and its failure: a parametric-bootstrap obstruction test with a planted positive ($z = +220$ against $+6.5$ elsewhere), a block-level dependence measure, and a β_1 test for non-tree structure that fires on a local-optimum fit (6 of 25 thresholds), is silent on the recovered one (0 of 25), and catches planted couplings in both cases (9 of 25). Every structural identity used is verified numerically to 10^{-14} . We also record what does not work, including a probe result we withdraw, the vacuity of the cohomological gluing obstruction on a single tree, and a proof that the proposed p -adic lattice quasi-morphism is constant.

Contents

1	Setup: the generative model and its exact floors	2
2	Part I: transformers do not recover levels 3–4	3
2.1	Protocol	3
2.2	Result	3
2.3	Statement	4
3	Part II: the constructed estimator and its verification	4
3.1	Construction	4
3.2	Verification suite	4
3.3	Estimator variants	5
4	Part III: instruments for detecting hierarchy	5
4.1	V6: where does the tree fail (token resolution)	5
4.2	V7: dependence between hierarchy units	6
4.3	β_1 : is the block structure a tree at all	6
5	Part IV: the regime determines whether anything is learnable	6

6	Part V: what is not established	7
7	Closing the gap: the failure was basin selection	8
7.1	Diagnosis: start EM at the truth	8
7.2	Why the hard clustering cannot work	9
7.3	Split-merge: letting EM do the clustering	9
7.4	Verification of the recovered model	9
7.5	The gap to the transformer	10
8	Why clustering is necessary at scale	10
8.1	Cost of the dense recursion	10
8.2	Cost ladder for m -ary grammars	10
8.3	Clustering belongs inside EM, not before it	11
8.4	Compression is not free: the measured trade-off	11
8.5	What is projected and what is demonstrated	11
9	Open problems	12
10	Inventory of theorems and constructions used	13
11	Reproduction: commands and the result each produced	15

1 Setup: the generative model and its exact floors

All experiments use the Random Hierarchy Model (RHM) of Cagnetta et al. [9]: a probabilistic context-free grammar on a fixed complete binary tree of depth L , with V symbols per internal level, m productions per symbol, and a leaf alphabet of size A . A sequence is the string of 2^L leaves obtained by sampling a root symbol uniformly and then one production per node. Because the tree shape is fixed and positional, the latent structure is a Markov chain

$$\text{leaves} \rightarrow \text{level } 1 \rightarrow \cdots \rightarrow \text{root}, \quad (1)$$

and every conditional independence we use below is an instance of the tree Markov property.

Two corpora appear. Corpus A ($L = 4$, $V = 64$, $A = 48$, $m = 12$, 2×10^5 sequences of length 16) is the configuration inherited from earlier phases. Corpus B ($L = 4$, $V = 8$, $A = 12$, $m = 3$, 2×10^5 sequences) was built for this study; Section 5 explains why.

Positions are labelled *local* (odd indices, determined by the sibling) and *higher* (even indices, which open a new subtree). We compute the *exact* per-position conditional entropy $H(x_t | x_{<t})$ from the grammar by the inside–outside algorithm (Theorem 2), rather than estimating it. For Corpus A the means are

$$H_{\text{local}} = 1.6034 \text{ nats}, \quad H_{\text{higher}} = 3.5527 \text{ nats}, \quad (2)$$

with $H(x_0) = 3.8161$ alone (the first token of a fresh tree is nearly unpredictable), and 3.5150 for the higher positions excluding $t = 0$. These are floors: no predictor of any kind can go below them.

We also compute, exactly, the conditional entropy of each *latent* symbol given the observed prefix. Over the higher positions of Corpus A, against a chance value of $\ln V = 4.1589$:

level	1	2	3	4
$H(\text{symbol} x_{<t})$	2.7835	2.2923	2.2900	2.7910
fraction of chance	67%	55%	55%	67%

At individual positions the level-3 symbol is nearly determined (0.427 nats at $t = 14$, 1.126 at $t = 12$). This matters for what follows: roughly 1.87 nats of level-3 information are recoverable from the prefix, so a model that fails to represent it is failing at something the data support, not attempting the impossible.

2 Part I: transformers do not recover levels 3–4

2.1 Protocol

Decoder-only transformers were trained on Corpus A with next-token cross-entropy for 48,000 steps. After training, a linear probe was fitted from the residual stream $h[t]$ to the level- ℓ symbol above position t , with a 70/30 split over sequences and two controls: shuffled labels, and an untrained network of identical shape. Chance is $1/V = 0.0156$.

2.2 Result

configuration	local	higher	level-3 probe
2 layers, $d = 64$ (baseline)	2.173	3.831	0.028
4 layers	2.078	3.797	—
8 layers, frozen embedding	2.037	3.755	—
8 layers, frozen emb., fresh sampling	2.022	3.772	0.027
$d = 128$, 4 layers	2.039	3.770	—
frozen random embedding (2 layers)	2.186	3.824	—
constructed order coordinate	2.162	3.815	—
auxiliary supervision, levels 3–4	2.088	3.777	0.056
latent marginalising head (stopped at 34k)	3.227	3.835	—
dilated causal CNN (tree receptive fields)	2.104	3.827	0.032
curriculum over tree depth	2.085	3.803	0.034
curriculum control (no easy cases)	2.043	3.789	0.033
exact floor	1.603	3.553	—

Shuffled-label controls ran at 0.021–0.034; the level-3 probe is indistinguishable from them in every configuration. Level 4 read 0.016–0.020 against chance 0.0156. The auxiliary arm is the strongest test: its head was trained jointly for 48,000 steps on *ground-truth* level-3 and level-4 labels and reached 0.056 and 0.019.

Three additional controls sharpen the claim.

The embedding is not the site. With the token embedding frozen at random initialisation and never updated, final loss is unchanged (2.185/3.824 versus 2.163/3.831), and the sublayer readout shows the order structure being constructed entirely in the blocks: the embedding tap reads exactly 0.0000, then +0.0430 (attn₀), +0.0508 (mlp₀), +0.0410 (attn₁), +0.1272 (mlp₁).

A linear probe on the embedding was not evidence. An earlier phase reported a linear probe reading comparability off the embedding matrix at 0.98–1.00 on held-out multi-hop pairs. Running the identical probe on (i) a fresh random initialisation, (ii) an exactly orthonormal matrix, and (iii) the trained matrix with its *rows permuted* gives 1.000, 0.984 and 1.000. With $A + 1$ near-orthogonal vectors in $d = 64$, any scalar ranking of tokens is linearly realisable, so the probe carries the order in its own weights. That result is withdrawn; the hidden-state measurements, which use no probe, are unaffected.

The one direction that is causal is causal for local. Projecting the fitted order direction out of the embedding, against a random-direction control at matched rank, costs an excess of +0.079 to +0.100 nats on the overall loss (4.3–6.2 control standard deviations) — of which **local** absorbs +0.279 and **higher** +0.0035.

2.3 Statement

Across twelve configurations spanning depth, width, input representation, output factorisation, architecture family, objective and curriculum, the level-3 probe never leaves the shuffled-label floor, while 1.87 nats of level-3 information are present in the prefix. We do not claim impossibility; recurrent-depth architectures were tried only briefly, and the mechanistic account below is a hypothesis.

Hypothesis (not established). Recovering the level- ℓ symbol requires marginalising over parses consistent with the prefix. Inside–outside does this with an upward *and* a downward pass whose messages are coupled; a feedforward stack has only the upward one and must approximate the composite in a fixed number of layers. Consistently, depth from 2 to 8 buys 0.06 nats and no change in level-3 decodability.

3 Part II: the constructed estimator and its verification

3.1 Construction

No gradients are used. The rule tables $P_\ell[p, x, y]$ are estimated by:

1. **Spectral initialisation.** At level ℓ , two observed child pairs belong to the same parent if they occur in the same contexts. We build the pair–context co-occurrence matrix, take its leading singular directions, and k -means the pairs into V groups. Where a level has fewer than four nodes per sequence there is no horizontal context and we fall back to grouping by shared child identity.
2. **Soft hand-off.** The posterior over parent symbols, not a hard label, is propagated to the next level: $q_\ell \propto \sum_{x,y} P_\ell[\cdot, x, y] q_{\ell-1}^{\text{lo}}(x) q_{\ell-1}^{\text{hi}}(y)$.
3. **EM.** Inside–outside E-step, normalise-expected-counts M-step, iterated; the sweep with the best held-out likelihood is kept.

3.2 Verification suite

Each item below is an identity of the tree model class. Passing verifies the implementation and that the fitted object is a genuine section; it does *not* test whether the data are tree-generated. Errors are maxima over 256 held-out sequences of Corpus A.

check	statement	max error
V0	EM never decreases the training log-likelihood	$+1.8 \times 10^{-2}$ (min step)
V1a	a free subtree has inside value exactly 1	1.8×10^{-15}
V1b	restriction = marginalisation	8.9×10^{-16}
V1c	gluing = independence join	1.0×10^{-14}
V2	the hand-off is a sufficient statistic	6.7×10^{-16}
V4a	merge cost = $N I(\text{children}; B \mid q(B))$	1.2×10^{-14}

V1c is the concrete content of Theorem 6 and Construction 8: with $\mu(u, v)$ the joint prior on the two top children,

$$P(x_{0:16}) = \sum_{u,v} \mu(u, v) \beta_L(u) \beta_R(v), \quad (3)$$

which we verify numerically rather than assume. Note μ is the *joint*, not a product: the two halves are dependent through the root, and factorising them breaks the identity.

V2 is the operational form of Definition 7: replacing each completed subtree by its normalised inside vector and deleting its leaves changes the predictive distribution by $< 10^{-15}$.

V4a is Construction 11: merging symbols into their mixture row costs exactly $\sum_s n_s \text{KL}(R_s \| R_{\text{class}})$, which is the Bregman information of the merge and equals $N I(\text{children}; B \mid q(B))$ — zero precisely when the merge is lossless.

3.3 Estimator variants

On Corpus A, 2×10^4 training sequences:

variant	local	higher	held-out log p/seq
joint EM from random init	2.566	3.831	−51.18
bottom-up, frequency-rank clusters	2.598	3.833	−51.45
spectral + soft hand-off + EM (10 sweeps)	2.227	3.802	−48.23
spectral + soft hand-off + EM (50 sweeps)	2.162	3.807	−47.75
stagewise + vertical re-clustering	2.240	3.802	−48.34
transformer, 48k steps	2.022	3.772	—
true grammar	1.599	3.553	−41.21

Initialisation dominates. On Corpus A the constructed estimator does not beat the trained transformer; Section 5 shows why, and Section 7 closes the gap entirely on an identifiable corpus.

4 Part III: instruments for detecting hierarchy

Three measurements, each with a calibration control and, where possible, a planted positive.

4.1 V6: where does the tree fail (token resolution)

For each pair of leaf positions we compare empirical mutual information in the data against the same statistic computed on datasets *sampled from the fitted model* at equal size (parametric bootstrap, 20 replicates). Because the same estimator is applied to both sides, plug-in bias cancels. Controls: the true grammar against its own samples (calibration), and the true grammar against the data (which also verifies that the exported rule tables are the actual generator).

Planted positive. Copying leaf 7 into leaf 8 with probability 0.3 creates a dependence across the root that no tree of this shape carries. The detector reports $z = +220$ at the planted pair against a largest other z of +6.5; even the true grammar is refuted there ($z = +390$). During training, the data-minus-model gap at the planted pair stays at +0.600 across every sweep, while the sibling gap closes from +0.161 to +0.028: training absorbs structure the tree can hold and never touches structure it cannot.

Limitation. On Corpus A, pairwise leaf MI across level-2 and root boundaries is $\approx 10^{-4}$ even under the true grammar, while the plug-in bias floor at 4×10^4 sequences is ≈ 0.055 . V6 therefore has no power above level 1 there; the 0.3 copy was detectable only because it was gross.

4.2 V7: dependence between hierarchy units

The fix is to test blocks, not tokens. Each block of 2^ℓ leaves is decoded to the level- ℓ symbol given by the argmax of *that block's own* inside vector, and mutual information is measured between adjacent decoded blocks, with the same parametric bootstrap.

Disjointness invariant. Every statistic compares functions of disjoint leaf sets. Blocks share no leaves, and decoding uses inside messages only: outside messages would import the neighbour's evidence and make adjacent blocks appear dependent under any model. Outside messages remain legitimate for *fitting*; they are excluded only from the test decoder.

Result (Corpus A, 2×10^4 sequences). Decoded with the true grammar, blocks sharing a parent carry ≈ 1.0 nat at levels 1, 2 and 3, and blocks not sharing one carry 0.002–0.014. The fitted model captures $0.866/1.102 = 79\%$ at level 1, $0.014/0.994 = 1.4\%$ at level 2 and $0.038/0.965 = 3.9\%$ at level 3. All calibration controls pass ($|z| \leq 2.6$).

4.3 β_1 : is the block structure a tree at all

Decode blocks as in V7 and build the graph on blocks whose pairwise MI exceeds a threshold. Under a tree the dependency structure is nested: every threshold cuts the graph into cliques (the blocks under one ancestor), and a clique complex of cliques has no holes. Hence the prediction

$$\text{tree} \implies \beta_1 = 0 \text{ at every threshold,} \quad (4)$$

with $\beta_1 = \dim \ker \partial_1 - \text{rank } \partial_2 = (E - V + C) - \text{rank } \partial_2$ computed exactly over $\text{GF}(2)$ on the 2-skeleton. A null band is obtained from datasets sampled from the fitted tree.

Result and planted positive (Corpus B). Unplanted: $\beta_1 = 0$ at all 25 thresholds, null band 0–1. Copying block 0 onto block 7 and block 3 onto block 4, each with probability 0.5: $\beta_1 = 1$ at five thresholds with the null at 0.

Two facts about the test. First, a *single* extra coupling produces no cycle: it is a bridge between two cliques, and β_1 counts independent loops, so at least two couplings between distinct clique pairs are required. Second, planting must be done at block level. Copying a single leaf creates ungrammatical pairs whose inside vector is numerically zero, the decoded symbol degenerates, and the coupling disappears; copying whole blocks keeps every production valid.

5 Part IV: the regime determines whether anything is learnable

The failure of the estimator at levels 2–4 on Corpus A is an identifiability problem, not an algorithmic one. Level ℓ has $2^{L-\ell}$ nodes per sequence and a table of $V \cdot c_\ell^2$ entries:

	level 1	level 2	level 3	level 4
Corpus A: observations per parameter	1.4	0.31	0.15	0.08
Corpus B: observations per parameter	1389	1563	781	391

Corpus B ($V = 8$, $A = 12$, $m = 3$) was constructed so that every level is estimable. On it the same estimator, with no algorithmic change, recovers the hierarchy. Writing c for the block dependence the fitted model’s own symbols capture and g for the dependence its samples generate, and z for the bootstrap discrepancy:

level	30 sweeps			200 sweeps		
	c	g	z	c	g	z
1	0.4568	0.4574	−0.0	0.4568	0.4530	+1.7
2	0.3582	0.3595	−0.2	0.3804	0.3818	−0.2
3	0.0854	0.0507	+20.2	0.1827	0.1715	+5.4

Level 3 more than doubles between 30 and 200 sweeps and its discrepancy falls from +20.2 to +5.4; held-out log-likelihood improves from −17.97 to −17.51 (floor −15.32) and was still descending at sweep 200. The earlier failure was therefore (i) an unestimable regime and (ii) undertraining, in that order.

The symbols are not redundant. The merge sweep (V4) on Corpus B: at levels 1–3 every merge costs real conditional MI (0.142, 0.071, 0.128 nats/node at the first step) and held-out likelihood falls monotonically. Level 4 is the exception: $8 \rightarrow 6$ costs 0.058 nats/node with held-out change +0.0000, and $8 \rightarrow 2$ gives +0.0039. The root alphabet is genuinely redundant; the others are not.

A defect, later resolved. At token resolution the fitted model still under-produced cross-boundary dependence: 0.174 against a true 0.282 at the level-1 boundary, 0.0006 against 0.0035 at the root, unchanged from 30 to 200 sweeps and unchanged when the training set was quadrupled from 5×10^4 to 2×10^5 sequences (−17.5094 against −17.5113 held-out). It is resolved in Section 7.

6 Part V: what is not established

We record failed constructions, because several are attractive and the reasons they fail are cheap to check.

The embedding probe (withdrawn). See Section 2: a random matrix supports the same probe accuracy.

Cohomological obstruction to gluing is vacuous here. For a single joint distribution on a fixed tree, every family of marginals extends, by Theorem 5. The Čech obstruction of Abramsky–Mansfield–Barbosa [2] is therefore identically zero. It becomes meaningful only for genuinely overlapping contexts — several parse shapes over the same tokens, or models fitted on different corpora sharing variables. The β_1 test of Section 4 is the non-vacuous substitute.

Information cohomology computes entropy, not obstructions. Vigneaux [26], following Baudot–Bennequin [6], shows H^1 of the information site is generated by Shannon entropy. This characterises which functionals exist; it does not measure failure to glue. A proposed test — verify the chain rule $H(X, Y) = H(X) + H(Y \mid X)$ at each node — cannot fail when all three terms are computed from the same empirical joint: it is an algebraic identity.

The lattice quasi-morphism does not exist.

Proposition 1. *Let $\rho : \mathbb{Q} \rightarrow L$ be order-preserving for the real order and constant on each p -adic ball. Then ρ is constant.*

Proof. A p -adic ball $B(a, p^{-n}) \cap \mathbb{Q}$ contains $a + p^n m/k$ for all $m \in \mathbb{Z}$ and all k coprime to p , hence is dense in $\mathbb{R}$. Let B_1, B_2 be disjoint balls with $\rho(B_1) = c_1, \rho(B_2) = c_2$. By density choose $x < y < z$ with $x, z \in B_1$ and $y \in B_2$. Monotonicity gives $c_1 \leq c_2 \leq c_1$, so $c_1 = c_2$. $\square$

The underlying obstruction is Theorem 13: $\mathbb{R}$ and $\mathbb{Q}_p$ are inequivalent completions of $\mathbb{Q}$, and the p -adic norm of a probability measures divisibility, not magnitude ($|0.3|_3 = 1/3 < |0.5|_3 = 1$). Real-valued rounding to the grid $p^{-n}\mathbb{Z}$ is a genuine quasi-morphism with defect $\leq p^{-n}$, but retains nothing p -adic.

Posterior entropy is the wrong confidence criterion. After stagewise fitting on Corpus A the per-level posterior entropies are 0.198, 0.353, 0.541, 1.003 nats against a maximum of 4.159 — all low — while V7 shows those levels capturing under 4% of the available dependence. The model is confident about labels that carry no structure. The criterion that does separate them is $I(\text{parent}; \text{children})$ computed from the rule table, which needs no ground truth and reads 10.1, 8.1, 4.5, 5.3 against true values 8.9, 9.0, 9.0, 9.0.

Curriculum does not transfer. The curriculum result of Qiu et al. [21] (their ablation: 0.0% baseline, 1.6% with depth, 8.7% with depth and reversal, 99.9% with sample weighting) does not reproduce here. A matched control with no easy cases is marginally *better* on both components (2.043/3.789 versus 2.085/3.803), and level-3 decoding is unchanged. Their failure mode was a broken easy component; ours is not.

7 Closing the gap: the failure was basin selection

7.1 Diagnosis: start EM at the truth

The shortfall of Section 5 admits three explanations: the model class is wrong, the data are insufficient, or the optimiser is in a bad basin. The third is testable directly. Initialise EM at the true rule tables, optionally blended with uniform at weight ε , and run it.

start	held-out log p /seq	local / higher
exact truth ($\varepsilon = 0$)	−15.324	0.3322/1.5833
truth + 50% uniform	−15.325	0.3322/1.5834
truth + 80% uniform	−15.325	0.3322/1.5834
truth + 90% uniform	−15.330	—
truth + 95% uniform	−15.775	—
truth + 99% uniform	−18.113	—
uniform ($\varepsilon = 1$)	−20.706	0.5091/2.0791
spectral init (ours)	−17.511	0.3952/1.7937

Three conclusions. The truth is a fixed point, so the model class is right and the corpus is what we believe it to be. The basin is very wide: a start carrying only 10% of the truth, whose likelihood (−31.2) is worse than uniform, climbs all the way back. And our spectral initialisation performs like a 1% admixture — it fails to clear a low bar.

7.2 Why the hard clustering cannot work

The initialisation assigns each observed child pair to exactly one parent cluster. Real grammars share pairs across parents: in Corpus B, 4 of the 16 distinct level-1 pairs are generated by two different symbols (2 of 18 at level 2, 4 of 16 at levels 3 and 4). A hard partition therefore cannot represent a quarter of the grammar however good the clustering is. Two predictions follow, and both hold: sharpening the clustering makes matters worse (using the sibling co-parent as context instead of linear adjacency gives -18.06), and blurring it helps (blending with uniform at weight 0.9 gives -15.87).

7.3 Split-merge: letting EM do the clustering

The remaining gap is closed by moving the clustering *inside* EM. For each level in turn:

1. **Split:** duplicate every level- ℓ symbol with a multiplicative perturbation, doubling that level’s alphabet; the child axes of level $\ell + 1$ double with it.
2. **EM:** re-fit the enlarged model.
3. **Merge:** greedily collapse back to V symbols, cheapest first, where the cost of a merge is the Bregman information of Construction 11. The merge ranges over all $2V$ symbols, so it can recombine across the split rather than undo it. This is the escape mechanism.
4. Accept the round only if held-out likelihood improves.

stage	held-out $\log p/\text{seq}$	merge cost (nats/seq)
warmup, 40 EM sweeps	-17.187	—
round 1, levels 1–4	-15.491	0.160, 0.113, 0.008, 0.003
round 2, levels 1–4	-15.325	0.042, 0.011, 0.007, 0.005
round 3, levels 1–4	-15.3253	0.052, 0.020, 0.007, 0.004
true grammar	-15.3243	—

Total time 131 s. The final model is 0.001 nats per sequence from the truth, and **local/higher** read 0.3322/1.5835 against the true 0.3322/1.5833. One round was rejected (round 2, level 4), so the accept test is doing work.

7.4 Verification of the recovered model

Likelihood agreement is not by itself recovery of structure. All four instruments confirm it independently.

	before split-merge	after	true
V3, sibling MI	1.5445	1.5507	1.5523
V3, level-1 boundary	0.1739	0.2783	0.2819
V3, level-2 boundary	0.0153	0.0334	0.0331
V3, root boundary	0.0006	0.0034	0.0035
V6, boundaries flagged	11 of 18	0 of 18	—
V6, largest $ z $	+58.6	+1.7	—
V7, level 3 (c vs g)	0.1798/0.1720	0.5863/0.5882	0.6140
β_1 vs fitted null	6 of 25	0 of 25	—

The β_1 result is the sharpest. Section 4 reported 6 of 25 thresholds flagged against a null sampled from the fitted model, all in the weak-edge window where the fit under-produced dependence, while the same test against a null sampled from the *true* grammar gave 0 of 25 and still caught a planted violation at 9 of 25. The diagnosis was therefore that β_1 was measuring the basin, not the data. With the recovered model the flags vanish. The instrument fires on a bad fit, is silent on a good one, and detects planted non-tree structure in both cases.

7.5 The gap to the transformer

On Corpus A the exact floors are `local` = 1.599, `higher` = 3.553 and the best transformer of Section 2 reaches 2.022/3.772, with the level-3 latent symbol never decodable above the shuffled-label floor. On Corpus B, where every level is identifiable, the gradient-free estimator reaches 0.3322/1.5835 against floors of 0.3322/1.5833 — that is, the floor to three decimal places — in 131 seconds without computing a single gradient. We emphasise that these are different corpora and the comparison is therefore about *kind*, not a head-to-head benchmark: the transformer fails to represent the upper levels at all under twelve configurations, whereas the estimator recovers them exactly when the regime is identifiable, and its remaining failures are diagnosable (basin selection) and curable (split-merge) rather than opaque.

8 Why clustering is necessary at scale

Sections 3–5 used $V = 8$, where a dense rule tensor is trivial. The motivation for cluster primes is what happens at $V = 10^3$.

8.1 Cost of the dense recursion

Let N be the total number of leaf tokens in the corpus. A complete binary tree over N leaves has $N - 1$ internal nodes *in total*: the levels partition the nodes, they do not replicate them. Each node evaluates

$$T_\ell(b, n, p) = \sum_{x, y} P_\ell(p \rightarrow x, y) T_{\ell-1}(b, 2n, x) T_{\ell-1}(b, 2n+1, y), \quad (5)$$

at a cost of $2V^3$ operations, so one inside pass costs $\approx 2NV^3$. With outside and the M-step, a sweep is $\approx 4NV^3$. At $N = 10^6$ and $V = 1024$:

$$C_{\text{sweep}}^{\text{dense}} \approx 4.3 \times 10^{15}, \quad C_{50 \text{ sweeps}}^{\text{dense}} \approx 2.1 \times 10^{17}. \quad (6)$$

Memory splits into two parts that scale differently. The inside table is $\approx 2NV$ floats, 8 GB, which is large but not fatal; the rule tensors are V^3 entries *per level*, 4.3 GB each and 69 GB over $L = 16$, which is. Replacing V by K gives $4NK^3$ per sweep, $\approx 8 \times 10^{11}$ over 50 sweeps at $K = 16$, with rule tensors of 16 KB per level, and a speedup of

$$\left(\frac{V}{K}\right)^3 = 64^3 = 262,144. \quad (7)$$

8.2 Cost ladder for m -ary grammars

For an m -ary grammar the rule tensor has order $m + 1$ and the per-node cost is V^{m+1} dense, K^{m+1} compressed. Each extra tensor order multiplies the compressed cost by K and the dense cost by V , so the compression buys $\log_K V$ orders of headroom — at $V/K = 64$, about 2.5 orders.

arity m	tensor order	dense ($V = 1024$)	compressed ($K = 16$)
2 (binary)	3	2.1×10^{17}	8×10^{11}
3 (ternary)	4	2.2×10^{20}	1.3×10^{13}
4	5	2.2×10^{23}	2.1×10^{14}
5	6	2.3×10^{26}	3.4×10^{15}

8.3 Clustering belongs inside EM, not before it

Section 7 revises what the initial clustering is for. An externally imposed partition supplies a seed and nothing more: sharpening it hurts, blurring it helps, and the symbols that finally matter are produced by merge moves *inside* EM, scored by the conditional mutual information of Construction 11 on expected counts from the E-step. The scaling argument above is unaffected — at $V = 10^3$ the V^3 contraction must be compressed — but the compression should be discovered by split-merge rather than fixed in advance.

8.4 Compression is not free: the measured trade-off

The scaling argument alone would suggest taking K as small as convenient. The merge sweep of Section 5 measures what that costs. On Corpus B, per level, I/node is the conditional mutual information paid (Construction 11) and Δ is the change in held-out log-likelihood per sequence:

K	level 1		level 2		level 3		level 4	
	I/node	Δ	I/node	Δ	I/node	Δ	I/node	Δ
8	0.000	+0.000	0.000	+0.000	0.000	+0.000	0.000	+0.000
6	0.142	+0.000	0.071	−0.222	0.128	−0.018	0.058	+0.000
4	0.397	−0.970	0.321	−0.716	0.430	−0.121	0.326	+0.003
3	0.626	−1.502	0.559	−0.838	0.663	−0.122	0.489	+0.003
2	0.983	−1.957	0.941	−0.880	1.063	−0.124	0.723	+0.004

Levels 1–3 pay for every merge in both currencies. Level 4 does not: $8 \rightarrow 2$ costs 0.72 nats per node of conditional MI and *improves* held-out likelihood by 0.004, because the root alphabet is redundant at this depth. So the right K is level-dependent and must be measured, not assumed; the merge identity V4a supplies the measurement.

8.5 What is projected and what is demonstrated

The recovery results of Section 5 are at $L = 4$, $V = 8$, 2×10^5 sequences. The costs above are arithmetic, not measurements: we have not run $L = 16$, $V = 1024$. One caveat is worth stating in advance. At $L = 16$ a sequence has $2^{16} = 65,536$ leaves, so $N = 10^6$ tokens is about 15 sequences, and the root level would be observed 15 times. The compression removes the compute barrier; it does not remove the observations-per-parameter barrier of Section 5, which at the top of a deep tree is governed by the number of *sequences*, not tokens.

A proposed argument we do not use. It is tempting to argue that attention is architecturally incapable of representing a rank- m rule tensor because each head is a rank-2 bilinear form and CP rank $\Theta(V^{m-1})$ would require V^{m-1} heads. We do not make this claim. The network computes a predictive function, not the rule tensor; the MLP sublayers, which our own readout shows carrying most of the order construction (+0.127 at mlp_1 against +0.041 at attn_1), are absent from the argument; and depth composes heads nonlinearly. The transformer result of Section 2 is an empirical negative over twelve configurations and does not need it.

9 Open problems

All of these concern the estimator or the instruments. None concerns cohomology: Section 6 records why the gluing obstruction is vacuous for a single distribution on a fixed tree, and nothing since has changed that.

O0. The cross-boundary defect, and why it is no longer open. Until Section 7 the binding problem was that the fitted model under-produced leaf-pair dependence at the level-1 and level-2 boundaries (0.174 against a true 0.282), and the natural reading was that the many-to-one block decoder cannot force the induced leaf marginals to match. That reading was wrong: the defect was an artefact of the basin, and it disappears entirely once split-merge finds the right optimum. We record it because the wrong reading was plausible and survived several rounds of measurement. It remains untested on Corpus A, where split-merge has not been run.

O1. A regime-identifiability threshold. We measure that levels fail at 0.15 observations per rule-table parameter and succeed at 781, but we have no threshold $\theta(V, L, \ell, m)$ separating them. What is wanted is a statement of the form: with n sequences of a depth- L , V -symbol, m -rule grammar, the level- ℓ table is identifiable up to relabelling with probability $1 - \delta$ provided $n \cdot 2^{L-\ell} \geq \theta$. The sample complexity of latent-tree models is known in the spectral setting [10]; the EM setting with our initialisation is not covered.

O2. When does redundancy justify collapsing a level? Construction 11 gives the exact cost of a merge, and Section 5 shows level 4 collapsing from 8 to 2 symbols at a held-out *gain*. What is missing is a rule: given the merge cost $N I(\text{children}; B \mid q(B))$ and the parameter count released, predict the sign of the held-out change without measuring it. This is a model-selection statement—the natural form is a comparison of the merge cost against the effective number of parameters removed—and would turn the K -sweep from a search into a calculation.

O3. Why rule-table mutual information beats posterior entropy. Section 6 shows posterior entropy $H(q_\ell)$ is low (0.20–1.00 nats against a maximum of 4.16) at levels that capture under 4% of the available dependence, while $I(\text{parent}; \text{children})$ computed from the rule table separates them (10.1, 8.1, 4.5, 5.3 against true 8.9, 9.0, 9.0, 9.0). The observation is that a model can be certain about labels that carry no structure. We have no theorem saying when the two diverge, nor a threshold on $I(\text{parent}; \text{children})$ that certifies a level as determined — which is what a stopping rule for stagewise fitting needs.

O4. Power of the β_1 test. The detector finds two planted block couplings at 9 of 25 thresholds with the null at 0, and a single coupling never, because one extra edge is a bridge rather than a loop. What is the detection threshold as a function of the number of couplings m , their strength, the number of blocks and the sample size? At minimum, $m \geq 2$ is necessary; the sufficient condition and the power curve in coupling strength are unmeasured.

O5. A genuinely non-tree corpus. Every positive control here is a *planted* coupling made by copying one block onto another, which is destructive: it adds a dependence and removes the one it overwrites. At level 2, with only four blocks, this makes a 4-cycle unreachable — every plant we tried broke one of the four edges it needed. The clean test is a corpus *generated* from a non-tree structure, so that the detector can be scored against a known ground truth rather than a perturbation. This is a corpus-builder change, not a detector change.

O6. Split-merge at scale. The recovery result is at $L = 4$, $V = 8$, 2×10^4 sequences, in 131 seconds. Section 8 gives the arithmetic for $L = 16$, $V = 10^3$ but no measurement. The specific risk named there is that compression removes the compute barrier and not the identifiability barrier: at $L = 16$ a sequence has 65,536 leaves, so 10^6 tokens is about 15 sequences and the root level is observed 15 times. Whether split-merge tolerates that, and whether the split/merge cycle stays stable when the alphabet is large enough that the doubled model does not fit in memory, are untested.

O7. Recurrent-depth transformers. Section 2 covers twelve configurations, all with distinct weights per layer. Weight sharing across depth — a single block applied L times — is the one architecture family in which iterated message passing is representable at fixed parameter count, and it was tried only briefly. A negative there would considerably strengthen the claim; a positive would locate what the feedforward stack is missing.

10 Inventory of theorems and constructions used

Used and load-bearing

Theorem 2 (Inside–outside; Baker [4], Lari–Young [16]). *For a PCFG on a fixed tree, the inside values β and outside values α computed by the two recursions give exact marginals: the posterior at a node is $\propto \alpha\beta$, and the marginal likelihood is the root inside value against the prior.*

Theorem 3 (EM ascent; Dempster–Laird–Rubin [12]). *Each EM iteration does not decrease the observed-data log-likelihood. The proof is Jensen’s inequality on the Q -function; no geometric hypothesis is needed.*

Theorem 4 (Convergence; Wu [29]). *Under regularity conditions the EM sequence converges to a stationary point of the likelihood; the limit depends on the initialisation.*

Theorem 5 (Extension; Vorob’ev [27]). *A family of consistent marginals on a hypergraph extends to a global distribution for every consistent family if and only if the hypergraph is acyclic. A fixed tree is acyclic.*

Theorem 6 (Independent pullback; Simpson [23], Thm. 6.4 and Prop. 8.5). *A presheaf on standard Borel probability spaces is an atomic sheaf if and only if it maps independent squares to pullbacks; the independent pullback of a cospan is the conditional product $\int P(\cdot \mid r=\omega) \otimes P(\cdot \mid s=\omega) dP(\omega)$.*

Definition 7 (Supports; Simpson [23], Def. 7.1). *A support of a random variable is the terminal factorisation of it through a sample space: the smallest randomness it depends on. $\text{RV}(A)$ has supports.*

Construction 8 (Independence join; Wright [28], §7.1). *For tables agreeing on a shared marginal, $(p \bowtie q)_{ijk} = p_{ij}q_{jk}/r_j$ is a weak pullback and is the maximum-entropy distribution with those marginals. This is the same object as Theorem 6 and is the content of (3).*

Construction 9 (Merging as convex combination; Wright [28], §7.2). *Conflicting categorical tables are merged by averaging distributions in the Giry monad. The merged rule row is the mixture of the rows merged.*

Theorem 10 (Bregman information; Bregman [7], Banerjee et al. [5]). *For the Bregman divergence generated by negative entropy — that is, for KL — the optimal representative of a cluster is its mean, and the clustering loss is the Bregman information, which equals the mutual information between the data and the cluster assignment.*

Construction 11 (Merge criterion). Merging latent symbols into their mixture row costs $\sum_s n_s \text{KL}(R_s \| R_{\text{class}}) = N I(\text{children}; B \mid q(B))$, zero exactly when the merge is lossless. Losslessness is strong lumpability in the sense of Kemeny–Snell [15]: the merged rows must be identical.

Construction 12 (Agglomerative merging; Slonim–Tishby [24]). Greedy merging of the pair with least information loss is the agglomerative information bottleneck; the split-merge PCFG estimator of Petrov et al. [20] is the grammar-specific analogue.

Theorem 13 (Ostrowski [19]). *Every nontrivial absolute value on $\mathbb{Q}$ is equivalent either to the real absolute value or to a p -adic one, and the resulting completions $\mathbb{R}, \mathbb{Q}_2, \mathbb{Q}_3, \dots$ are pairwise inequivalent.*

Construction 14 (Bias correction). Plug-in mutual information on a $K \times L$ table from N samples is biased upward by $\approx (K - 1)(L - 1)/2N$ (Miller–Madow [17]). The tracker applies the correction; the bootstrap does not need it, because the same estimator is applied to data and to model samples.

Construction 15 (β_1 of the clique complex). $\beta_1 = \dim \ker \partial_1 - \text{rank } \partial_2$, with $\dim \ker \partial_1 = E - V + C$ the cycle rank, computed over $\text{GF}(2)$. See (4).

Consulted, with the reason they are not used

- **Abramsky–Brandenburger [1], Abramsky–Mansfield–Barbosa [2]**: sheaf-theoretic contextuality and its Čech obstruction. Vacuous for a single distribution on one fixed tree (Theorem 5); relevant only for overlapping contexts.
- **Vigneaux [26], Baudot–Bennequin [6]**: information cohomology. H^1 is generated by entropy; this is a characterisation, not an obstruction theory.
- **Four-point condition / latent-tree distances (Buneman [8]; Choi et al. [10])**: the additive information distance assumes equal state counts and full-rank joints. With $A = 48$ leaves and $V = 64$ latents neither holds, so additivity is not guaranteed and the test would not be trustworthy. The parametric bootstrap is valid regardless of state counts.
- **Chow–Liu [11]**: gives the best tree over *observed* variables; our structure is latent.
- **Amari [3]**: e - and m -projections. The E-step is the e -projection and the M-step the m -projection; EM monotonicity follows from Theorem 3 and does not require the Pythagorean identity.
- **Tucker [25] / non-negative Tucker**: replacing P_ℓ by a rank- R core reduces V^3 parameters to $R^3 + 3VR$. Not needed at $V = 8$; becomes necessary at $V = 10^3$, where a dense table is 10^9 entries. Proposed, not implemented.
- **Ilić–Stepić et al. [14]**: conditional p -adic probability logic. Establishes that Monna–Springer boundedness needs the range restricted to $\mathbb{Z}_p$ or the propositional alphabet kept finite, and states explicitly that $\mathbb{Q}_p$ admits no compatible order — which is why Assumption 3 (monotone along the tree, a statement about the real order) and p -adic valuation cannot be imposed together.
- **Gauthier [13]**: recasts probability spaces as Grothendieck sites and then removes time and probability, so the absence of uncertainty is definitional rather than derived. No algorithm; nothing bearing on estimation.

- **Rute [22]**: computable measure theory and algorithmic randomness. Relevant only if a *constructive* convergence statement for EM were wanted in place of Theorem 4.
- **Ostrowski [19], Monna–Springer [18]**: used only to establish Proposition 1 and the incompatibility of p -adic valuation with the real ordering of probabilities.

11 Reproduction: commands and the result each produced

Every number in this section came from the commands below, in this order. Corpus A is the inherited configuration ($V = 64$, $A = 48$); Corpus B is the identifiable one ($V = 8$, $A = 12$).

Corpora

```
python3 build_corpus_syn.py --out /tmp --synonym --reverse-valid \
    --nleaf 48 --syn-frac 0.35 --nsym 64 --nrules 12 \
    --poset-density 0.25 --n-train 200000
```

Corpus A. Observations per rule-table parameter: 1.4, 0.31, 0.15, 0.08 at levels 1–4. Held-out bigram 3.534.

```
python3 build_corpus_syn.py --out /tmp/CH --depth 4 --nsym 8 --nrules 3 \
    --nleaf 12 --reverse-valid --n-train 200000 --n-val 5000
```

Corpus B. Observations per parameter: 1389, 1563, 781, 391. Exact entropy 1.160 nats/token.

Exact floors and latent-parse entropies

```
python3 floor_exact.py --data /tmp --samples 500
```

Corpus A: $H_{\text{local}} = 1.6034$, $H_{\text{higher}} = 3.5527$, $H(x_0) = 3.8161$. Latent-symbol entropy given the prefix, averaged over higher positions: 2.784, 2.292, 2.290, 2.791 against chance 4.159 — so 1.87 nats of level-3 information are recoverable.

Transformer arms (Corpus A, 48,000 steps each)

```
python3 rhm16.py --mlp --layers 2 --freeze-emb --steps 48000 --seed 1 --save L2.pt
python3 rhm16.py --mlp --layers 4 --freeze-emb --steps 48000 --seed 1 --save L4.pt
python3 rhm16.py --mlp --layers 8 --freeze-emb --steps 48000 --seed 1 --save L8.pt
python3 rhm16.py --mlp --layers 4 --d-model 128 --freeze-emb --steps 48000 \
    --seed 1 --save W128.pt
python3 rhm17.py --mlp --layers 8 --freeze-emb --steps 48000 --seed 1 --save A0.pt
python3 rhm17.py --mlp --layers 8 --freeze-emb --steps 48000 --seed 1 \
    --fresh-sample --save A1.pt
python3 rhm17.py --mlp --layers 8 --freeze-emb --steps 48000 --seed 1 \
    --aux-levels 3,4 --save A2.pt
python3 tree_cnn2.py --steps 48000 --seed 1 --save C2.pt
python3 curriculum.py --steps 48000 --seed 1 --save K1.pt
python3 curriculum.py --steps 48000 --seed 1 --simple-frac 0 --save K0.pt
python3 latent_head.py --steps 48000 --seed 1 --save M1.pt
python3 looped.py --loops 4 --steps 48000 --seed 1 --save P4.pt
```

local/higher at step 48,000, floors 1.603/3.553: 2.173/3.831 (2 layers), 2.078/3.797 (4), 2.037/3.755 (8 frozen), 2.022/3.772 (8 frozen, fresh sampling), 2.039/3.770 ($d=128$), 2.088/3.777 (auxiliary supervision), 2.104/3.827 (dilated CNN), 2.085/3.803 and 2.043/3.789 (curriculum and its control), 3.227/3.835 (latent head, stopped at 34k).

Probes and controls (Corpus A)

```
python3 parse_state.py --load A1.pt --layers 8
python3 parse_state.py --load A2.pt --layers 8
python3 null_probe.py --load s1.pt
python3 ablate_w.py --load s1.pt
```

Level- ℓ decoding from $h[t]$: 0.479/0.213 (level 1), 0.072/0.064 (2), 0.028/0.024 (3), 0.020/0.018 (4), against shuffled 0.021–0.034 and chance 0.0156. The auxiliary head trained on ground-truth labels reaches 0.056 (level 3) and 0.019 (level 4). `null_probe` returns 1.000 on a random matrix, 0.984 on an orthonormal one and 1.000 on the trained matrix with rows permuted: the embedding probe is retired. `ablate_w` gives an excess of +0.079–+0.100 nats over a matched random-direction control (4.3–6.2 sd), of which `local` absorbs +0.279 and `higher` +0.0035.

Estimator and verification suite

```
python3 estimator2.py --data /tmp --n-train 20000 --polish 50

python3 estimator4.py --data /tmp --n-train 20000 --polish 50 --skip-v4 \
    --boot 20 --boot-n 40000 --save fit20k.npz

python3 estimator4.py --data /tmp/cH --n-train 200000 --polish 200 \
    --boot 20 --boot-n 40000 --skip-v4 --save /tmp/cH/fit200k.npz
```

Corpus A: 2.162/3.807, held-out -47.75 (the numpy port of `estimator4` reproduces `estimator2` to 10^{-3}). Corpus B at 2×10^5 sequences: 0.3952/1.7937, held-out -17.511 — identical to the 5×10^4 run (-17.509), so the residual gap is not sample-limited. Verification: V1a 1.8×10^{-15} , V1b 8.9×10^{-16} , V1c 1.0×10^{-14} , V2 6.7×10^{-16} , V4a 1.2×10^{-14} .

```
python3 estimator4.py --data /tmp/cH --n-train 50000 --polish 30 \
    --boot 4 --boot-n 40000 --ks 8,6,4,3,2
```

Merge sweep: levels 1–3 pay in both currencies (first-merge I/node 0.142, 0.071, 0.128; held-out falls monotonically to $-1.96, -0.88, -0.12$ at $K = 2$). Level 4 is redundant: $8 \rightarrow 2$ costs 0.72 nats/node and *improves* held-out by +0.004.

Diagnosis of the local optimum

```
python3 estimator4.py --data /tmp/cH --n-train 20000 --polish 15 --init-true 0.0 \
    --boot 3 --boot-n 4000 --skip-v4
python3 estimator4.py --data /tmp/cH --n-train 20000 --polish 40 --init-true 0.9 \
    --boot 3 --boot-n 4000 --skip-v4
python3 estimator4.py --data /tmp/cH --n-train 20000 --polish 60 --anneal 5 \
    --boot 3 --boot-n 4000 --skip-v4
python3 estimator4.py --data /tmp/cH --n-train 20000 --polish 60 --sib-ctx \
    --boot 3 --boot-n 4000 --skip-v4
python3 estimator4.py --data /tmp/cH --n-train 20000 --polish 300 --init-mix 0.9 \
    --boot 3 --boot-n 4000 --skip-v4
```

Truth is a fixed point (-15.324 , unchanged over 15 sweeps). Truth blended with 90% uniform recovers (-15.330); with 99% it does not (-18.113). Annealing at $T_0 \in \{2, 5, 20\}$ all give -17.632 . Sibling context is *worse* (-18.06). Blending the init with uniform at 0.9 gives -15.865 .

Split-merge

```
python3 splitmerge.py --data /tmp/cH --n-train 20000 --warmup 40 --rounds 3 \  
    --split-sweeps 20 --post-sweeps 20 --save /tmp/cH/sm.npz
```

```
python3 estimator4.py --data /tmp/cH --load-P /tmp/cH/sm.npz \  
    --n-train 20000 --boot 20 --boot-n 40000 --skip-v4
```

Held-out -15.3253 against the true grammar's -15.3243 , in 131 s. $\text{local/higher} = 0.3322/1.5835$ against $0.3322/1.5833$. Verification of the recovered model: V3 matches the true column at every boundary, V6 flags 0 of 18 with largest $|z| = 1.7$, V7 tracks at all three levels with $|z| \leq 1.6$.

Obstruction and β_1 detectors

```
python3 estimator4.py --data /tmp/cH --n-train 50000 --polish 30 \  
    --plant 7,8,0.3 --boot 20 --boot-n 40000 --skip-v4
```

```
python3 betti.py --data /tmp/cH --load-P /tmp/cH/fit.npz --level 1 \  
    --n 40000 --boot 20 --null true --show-mi
```

```
python3 betti.py --data /tmp/cH --load-P /tmp/cH/fit.npz --level 1 \  
    --n 40000 --boot 20 --null true --plant-block "0,7,0.5;3,4,0.5"
```

```
python3 betti.py --data /tmp/cH --load-P /tmp/cH/fit.npz --level 1 \  
    --n 40000 --boot 20 --null fitted
```

```
python3 betti.py --data /tmp/cH --load-P /tmp/cH/sm.npz --level 1 \  
    --n 40000 --boot 20 --null fitted
```

Planted leaf copy: $z = +220$ at the planted boundary against $+6.5$ elsewhere; the true grammar is refuted there at $z = +390$. β_1 against a true-grammar null: 0 of 25 unplanted, 9 of 25 with two planted block couplings. Against a fitted null: 6 of 25 for the local-optimum fit, 0 of 25 for the split-merge fit.

Note on settings. `--boot` below about 10 makes the bootstrap standard deviation unreliable and the control checks will fail spuriously; `--boot-n` below about 4×10^4 makes the Miller–Madow correction dominate the 64×64 block statistics. The small values above appear only in diagnostic runs where V6/V7 are not being read.

References

- [1] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.* 13:113036, 2011.
- [2] S. Abramsky, S. Mansfield, and R. S. Barbosa. The cohomology of non-locality and contextuality. *EPTCS* 95:1–14, 2012.
- [3] S. Amari. *Information Geometry and Its Applications*. Springer, 2016.
- [4] J. K. Baker. Trainable grammars for speech recognition. *Speech Communication Papers, 97th Meeting of the ASA*, 1979.
- [5] A. Banerjee, S. Merugu, I. S. Dhillon, and J. Ghosh. Clustering with Bregman divergences. *JMLR* 6:1705–1749, 2005.
- [6] P. Baudot and D. Bennequin. The homological nature of entropy. *Entropy* 17(5):3253–3318, 2015.

- [7] L. M. Bregman. The relaxation method of finding the common point of convex sets. *USSR Comput. Math. Math. Phys.* 7(3):200–217, 1967.
- [8] P. Buneman. The recovery of trees from measures of dissimilarity. In *Mathematics in the Archaeological and Historical Sciences*, 387–395, 1971.
- [9] F. Cagnetta, L. Petrini, U. M. Tomasini, A. Favero, and M. Wyart. How deep neural networks learn compositional data: the Random Hierarchy Model. *Phys. Rev. X* 14:031001, 2024.
- [10] M. J. Choi, V. Y. F. Tan, A. Anandkumar, and A. S. Willsky. Learning latent tree graphical models. *JMLR* 12:1771–1812, 2011.
- [11] C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Trans. Inform. Theory* 14(3):462–467, 1968.
- [12] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *JRSS B* 39(1):1–38, 1977.
- [13] R. Gauthier. Algebraic stochastic calculus. [arXiv:1407.6784](#), 2014.
- [14] A. Ilić-Stepić, Z. Ognjanović, and N. Ikodinović. Conditional p -adic probability logic. *Int. J. Approx. Reasoning* 55:1843–1865, 2014.
- [15] J. G. Kemeny and J. L. Snell. *Finite Markov Chains*. Van Nostrand, 1960.
- [16] K. Lari and S. J. Young. The estimation of stochastic context-free grammars using the inside–outside algorithm. *Computer Speech and Language* 4(1):35–56, 1990.
- [17] G. A. Miller. Note on the bias of information estimates. In *Information Theory in Psychology*, 95–100, 1955.
- [18] A. Monna and T. Springer. Intégration non-archimédienne. *Indag. Math.* 25:634–653, 1963.
- [19] A. Ostrowski. Über einige Lösungen der Funktionalgleichung $\varphi(x)\varphi(y) = \varphi(xy)$. *Acta Math.* 41:271–284, 1916.
- [20] S. Petrov, L. Barrett, R. Thibaux, and D. Klein. Learning accurate, compact, and interpretable tree annotation. *ACL*, 2006.
- [21] X. Qiu et al. Dissecting multiplication in transformers. [arXiv:2407.15360](#), 2024.
- [22] J. Rute. On the close interaction between algorithmic randomness and constructive/computable measure theory. [arXiv:1812.03375](#), 2019.
- [23] A. Simpson. Equivalence and conditional independence in atomic sheaf logic. [arXiv:2405.11073](#), 2024.
- [24] N. Slonim and N. Tishby. Agglomerative information bottleneck. *NIPS*, 1999.
- [25] L. R. Tucker. Some mathematical notes on three-mode factor analysis. *Psychometrika* 31:279–311, 1966.
- [26] J. P. Vigneaux. *Topology of Statistical Systems: A Cohomological Approach to Information Theory*. PhD thesis, Université Paris Diderot, 2019.
- [27] N. N. Vorob’ev. Consistent families of measures and their extensions. *Theory Probab. Appl.* 7(2):147–163, 1962.

- [28] G. Wright. An algebraic perspective on computing with data. PhD thesis, 2023.
- [29] C. F. J. Wu. On the convergence properties of the EM algorithm. *Ann. Statist.* 11(1):95–103, 1983.